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Novel Oral Benzimidazoles Found to be Antagonists of Melanin Concentrating Hormone Receptor 1 (MCHR1)

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Abstract :

Melanin concentrating hormone (MCH) is an essential mediator of power homeostasis and performs position in several issues consisting of weight problems, pressure, depression and anxiety. The synthesis and organic evaluation of novel benzimidazole derivatives as MCHR1 antagonists are described. The in vivo evidence of principle for weight loss with a lead compound from this series is exemplified. Weight problems, a continual ailment associated with an imbalance among energy consumption and expenditure,1 is steadily growing into a worldwide pandemic affecting the lives of more than one thousand million humans worldwide it's miles emerging as a lead reason of morbidity and associated chance factors which includes dyslipidemia, type 2 diabetes, stroke, cardiovascular sickness, sure types of most cancers, osteoarthritis and sleep apnea.

Keywords : Melanin concentrating hormone, Benzimidazoles, Obesity

Introduction :

Many organic objectives for treating weight problems, inclusive of centrally modulated satiety and starvation regulating structures, were evaluated in literature. Among centrally performing targets, melanin concentrating hormone (MCH) and its receptor have been studied significantly.

The hormone MCH is a 19-amino acid cyclic neuropeptide expressed predominantly in the lateral hypothalamus and zona incerta,5 even though a few stages are also found in the course of the significant anxious gadget (CNS) [1]. Amassed proof has established that MCH is an essential mediator of energy homeostasis. The antagonists of MCH receptor (MCHR-1) have also tested efficacy in rat models of urinary incontinence. It is suggested that a single injection of MCH into the CNS stimulates meals consumption in rats and the continual administration leads to expanded body weight. Transgenic mice over expressing MCH gene are prone to insulin resistance and weight problems. The obese rodents, which includes ob/ob, db/db, and Ay/a mice show over expression of MCH mRNA whereas mice missing the gene encoding MCH are hypophagic, lean, and generally tend to preserve extended metabolic quotes.

Steady with this phenotype, genetically altered mice that lack the gene encoding MCH receptor hold multiplied metabolic costs and continue to be lean in spite of hyperphagia on a normal weight loss plan.

The collected data supports the therapeutic utility of MCHR1 antagonists in remedy of obesity.MCH also plays critical role in different CNS disorders and MCHR1 antagonists can be useful in remedy of illnesses together with melancholy and tension [2].

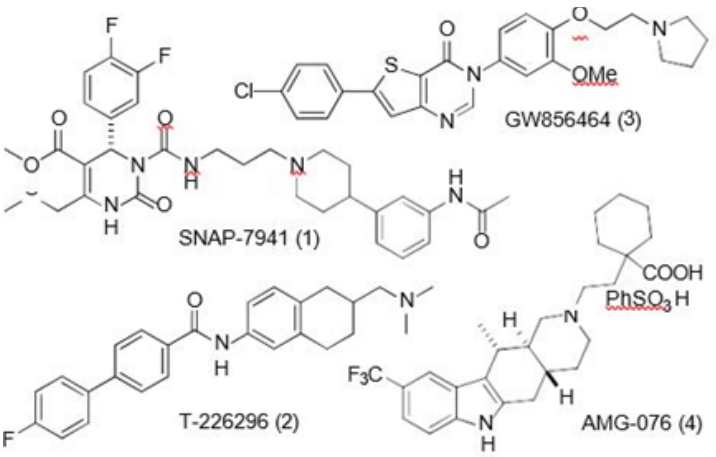


Figure 1: Potent MCHR1 antagonists.

The pioneering discovery of non-peptide MCHR1 antagonists SNAP-7941 (1, discern 1)14 and T-226296 (2)15 there had been giant efforts to develop MCHR1 antagonists16 as anti-weight problems sellers. More lately, GW856464 (3), AMG- 076 (4), and NGD-4715 (shape undisclosed) had been reported to enter phase I clinical trials. One among our laboratories has formerly reported selective and amazing MCHR1 antagonists derived from chemogenomics processes. In our efforts to are looking for novel MCHR1 antagonists with most suitable in vitro and in vivo profile, we have recognized and advanced benzimidazole derived series of compounds. Using SNAP-7941 as initial lead, systematic medicinal chemistry and computational procedures were used to increase proprietary collection with applicable physicochemical houses. This letter describes the synthesis and shape-interest research in generation of lead molecule with properly efficacy in obesity version [3].

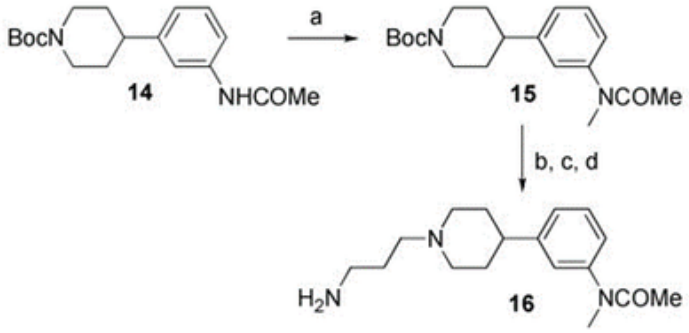
The target compounds had been synthesized. For the synthesis of compounds 22a-g, 23a-g and 24a-b, the benzimidazole element.

1. The SN2Ar substitution on 2-fluoronitrobenzene with various amines observed by reduction yielded diamines 5a-r. Heating with urea brought about formation of cyclic compounds 6a-r which had been sooner or later transformed to two-chlorobenzimidazoles 7a-r by treatment with POCl3. Compound 8 changed into prepared by means of oxidation of 7h with oxone.

Synthesis of the coupling companions 12a-d and 13a-c was carried. The compound 9 (prepared by the N-alkylation of 4-piperidinone with 2-(3- bromopropyl)isoindoline-1,3-dione) changed into converted to enol triflate 10 and matched with various boronic acids to yield eleven.

Selective manipulation and elimination of the phthalimide moiety22 with hydrazine supplied the corresponding amines [4]. The N-methylacetamide 16 changed into prepared from 14. In addition, synthesis of corresponding piperazine spinoff 21 became carried out from N-aryl piperazine 18 by alkylation with 19 observed through the typical differences protocol.

The coupling of the 2-chlorobenzimidazoles with various amines became finished in base mediated heating to the desired compounds.



F**igure 2**: Reagents and conditions.

Compounds 27a-c were prepared. Coupling of diamine (5g) with -chloro acid chloride afforded the corresponding 2-(ω-chloroalkyl)benzimidazoles 25 which after N-alkylation with piperidinylphenyl acetamide 26 (generated from 14) provided the targeted molecules 27a-c. In a comparable style, diamine 5g after treatment with methyl five-chloro-5- oxopentanoate observed by hydrolysis and coupling with 26 afforded the corresponding amide [5].

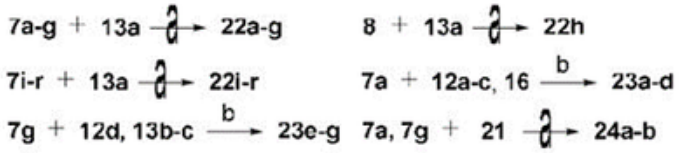


Figure 3: DBU 150°C, 1-4 h.

Synthesis of keto analog started with compound 30 which became prepared with the aid of copper-catalyzed N-arylation of benzimidazole following Buchwald's protocol. The remedy of 30 with LDA followed by using acylation with Weinreb amide (four-chloro-N-methoxy-N-methylbutanamide) afforded intermediate 31 which upon coupling with 26 produced 33 [6]. Compound 31 changed into handled with DAST to present 32 which became then converted to the corresponding difluoro analog 34.

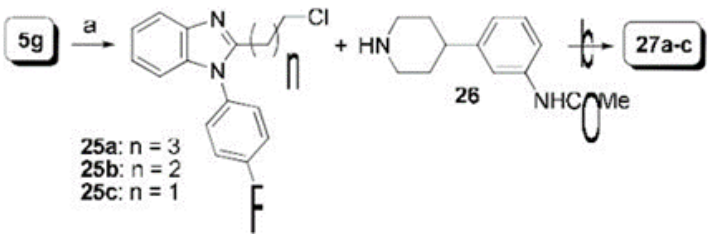


Figure 4: ω-chloro acid chloride, Et3N, CH2Cl2, rt, 1 h

The synthesis of amide 37. The diamine (5g) was dealt with with ethyl 2,2,2-triethoxyacetate to find the money for benzimidazole-2-carboxylic acid which changed into transformed to the corresponding acid chloride 35 and subsequently handled with amine 36 to furnish [7].

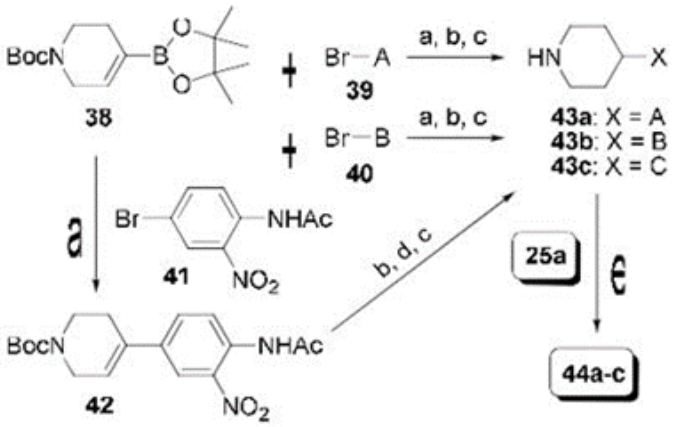


Figure 5: PdCl2(dppf), K2CO3, DMF, 80°C, 4 h

The compounds 44a-c, sporting acetanilide mimetics (viz., A, B and C, parent 2), were synthesized. The dioxaborolane derivative 38 changed into subjected to Suzuki coupling of with coupling companions within the presence of PdCl2(dppf). The artificial manipulations mentioned earlier supplied the critical arylpiperidine derivatives, which were N-alkylated with 25 to manage to pay for 44 [8].

The goal compounds were tested in a SPA based totally [125I] MCH binding assay the usage of chinese hamster ovary (CHO-K1) cellular membranes expressing human recombinant MCH1R receptors for his or her MCHR1 potency. The practical antagonism turned into measured in an IP3 SPA-YSI assay.

The consequences acquired by means of varying N- aryl and N-benzyl substituents in benzimidazole. In standard, para-substituents showed fairly higher efficiency over meta-substituents. The compounds with small substituents at para-position appear to furnish better potency. Adding larger functionality together with (SO2Me, OMe) display dramatic loss in activity.

Adding methylene spacer in 22a and 22g as in 22i and 22j respectively gave powerful compounds despite the fact that bulkier substituents at para- or meta-position endured to expose drop in potency [9]. The para-fluorophenyl by-product 22g displayed the first-rate efficiency in binding (IC50 2 nM) and efficacy in the useful assay (IC50 18 nM).

To apprehend the importance of acetamide group, compounds containing the desired para-fluoro- or para-chlorophenyl moieties. The decreased potency of compounds 23d and 23e vs 22a and 22g supports the need for both hydrogen bond donor and acceptor interactions on this area. The branched c-propyl and that i-propyl amides 23f and 23g, respectively, showed exact binding affinity despite the fact that reduced functional pastime as compared to 22g.

While the piperidine gadget become replaced with piperazine, dramatic loss in potency become discovered that is also likely related to the decreased basicity of piperidine nitrogen that appears to be involved in critical interaction with receptor [10].

Having explored western give up, we became interest to optimize linker chain. The replacement of NH moiety in 22g with a methylene unit gave 27a with stepped forward potency in each binding (IC50 0.6 nM) and purposeful (IC50 3 nM) assays.

The duration of the linker between japanese and western pharmacophoric motifs seems critical as molecules with reduced chain 27b and 27c showed > 50 fold loss in efficiency. The N-acylated analog confirmed > a thousand fold drop in potency (binding or purposeful assay) vs N-alkyl analog in addition assisting importance of basic piperidine nitrogen as mentioned above.

Variation of connector chain in close proximity to the benzimidazole turned into also investigated. The introdcution of carbonyl capability as in gave good binding affinity although drop in potency in functional assay. Converting the carbonyl organization to the corresponding CF2 analog progressed both binding and purposeful potencies on the other hand, carboxamide variation turned into negative to efficiency, in all likelihood related to its conformational influence [11].

Extra, practical mimetics of acetamide have been with one of the mightiest leads 27a. Compound 27a become docked into the main ligand binding pocket of the hMCHR1 receptor using a multi-conformational docking setup (described in the supplementary material).

The top scoring docking poses of 27a nicely fitted right into a complementary binding pocket of the hMCHR1 receptor in a low strength conformation which qualitatively is in good settlement with the overall SAR. In this docking pose, the coplanar acetamide moiety makes hydrogen bond interactions with a polar subpocket containing three glutamine residues in III:12, V:08 and VI:20 shown in discern three.35 further the posively charged piperidine nitrogen is forming an ionic interplay with aspartic acid III:08 and the 3- benzimidazole nitrogen paperwork a hydrogen bond with asparagines VII:02. The remaining scaffol makes true aromatic (part to stand and stacking) as well as hydrophobic interactions with a cluster of aromatic and hydrophobic residues between TM II, III and VII i.e. Ile II 21, Leu II:24, Met 25, Ile III:04, Tyr VII:02, Iso VII:06, Tyr VII:10 and aromatic residues in first extracellular loop.

Due to multiple rotamers and their viable engagement in energetically applicable binding modes with the glutamine residues, the prediction of favored acetamide mimetics changed into a challenging computational paintings. The analogs with rigid conformation or doubtlessly producing co-operative binding were explored. Those compounds (viz., salicylate (A), lactam (B) and imidazole (C); discern 2) when docked into the model produced reasonable interactions with at least of the 3 glutamine residues in TM III, V and VI. As the salicylate analog 44a become significantly less amazing vs the acetamide 27a. The two-methylbenzimidazole (44c) compatible with an acetamide rotamer conformation showed slight efficiency. It turned into gratifying to take a look at that the cyclic amide 44b displayed hMCHR1 potency akin to the lead acetamide by-product 27a with a binding affinity of one nM and useful antagonism of nine nM yielding a doubtlessly useful isosteric alternative [12].

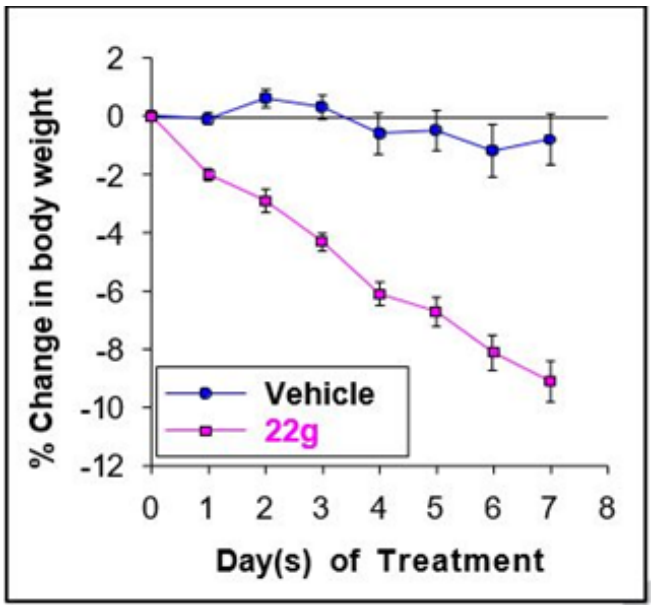


Figure 6: Effect of 22g on the body weight of DIO C57BL/6J mice. Mice were orally administered either vehicle or the MCHR1 antagonist at doses of 30 mg/kg, b.i.d. for 7 days.

As discussed above a number of fairly effective MCHR1 antagonists (including 22g, 27a, 34 and 44b) with low nanomolar efficiency in both binding and purposeful assay had been diagnosed. The lead compound 22g was evaluated similarly to set up proof of idea in a weight-reduction plan prompted weight problems (DIO) mouse model using C57BL/6J mice. On oral management 22g (30 mg/kg, b.i.d.), confirmed constant lack of frame weight culminating in a statistically tremendous weight reduction of 8% on day 7 [13].

In summary, we have discovered a benzimidazole derived novel collection of strong MCHR1 antagonists. Systematic structural hobby studies helped perceive key binding interaction and increase beneficial pharmacophore version. The consultant molecule 22g showed great anti-weight problems impact in a DIO mice model after sub-continual treatment for 7 days. Efforts are ongoing to further profile 22g and optimize collection and info can be disclosed in due course.

Conflict of Interest :

Author conforms there is no conflict of interest.

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